

PVsyst - Simulation report

Grid-Connected System

Project: Abhipsa Chakraborty

Variant: Project 4 22 Tilt -21.6 Orientation

No 3D scene defined, no shadings

System power: 30.2 kWp

Entrance Groningen - Netherlands

Author

Hanze university of applied sciences (Netherlands)



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Variant: Project 4 22 Tilt -21.6 Orientation

PVsyst V7.4.2

VC8, Simulation date:
17/10/23 17:02
with v7.4.2

Hanze university of applied sciences (Netherlands)

Project summary

Geographical Site

Entrance Groningen
Netherlands

Situation

Latitude 53.22 °N
Longitude 6.57 °E
Altitude 11 m
Time zone UTC+1

Project settings

Albedo 0.20

Meteo data

Eelde
Custom file - Imported

System summary

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Fixed plane
Tilt/Azimuth 18 / -24 °

Near Shadings

No Shadings

User's needs

Unlimited load (grid)

System information

PV Array

Nb. of modules 114 units
Pnom total 30.2 kWp

Inverters

Nb. of units 1 unit
Pnom total 25.00 kWac
Pnom ratio 1.208

Results summary

Produced Energy 32714 kWh/year Specific production 1083 kWh/kWp/year Perf. Ratio PR 86.94 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation

Orientation

Fixed plane

Tilt/Azimuth 18 / -24 °

Sheds configuration

No 3D scene defined

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

No Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer

REC

Model

REC 265PE

(Original PVsyst database)

Unit Nom. Power

265 Wp

Number of PV modules

114 units

Nominal (STC)

30.2 kWp

Optimizer Array

3 Strings x 19 In series

At operating cond. (50°C)

Pmpp

27.20 kWp

Output of optimizers

Voper

750 V

I at Poper

36 A

SolarEdge Power Optimizer

Model

P600 for SE15k+

Unit Nom. Power

600 W

Input modules

2 in series

Total PV power

Nominal (STC)

30 kWp

Total

114 modules

Module area

188 m²

Cell area

166 m²

Inverter

Manufacturer

SolarEdge

Model

SE25K-ISR

(Original PVsyst database)

Unit Nom. Power

25.0 kWac

Number of inverters

1 unit

Total power

25.0 kWac

Operating voltage

750 V

Pnom ratio (DC:AC)

1.07

Total inverter power

Total power

25 kWac

Number of inverters

1 unit

Pnom ratio

1.21

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const)

20.0 W/m²K

Uv (wind)

0.0 W/m²K/m/s

Module Quality Loss

Loss Fraction

-0.5 %

DC wiring losses

Global array res.

279 mΩ

Loss Fraction

1.5 % at STC

Module mismatch losses

Loss Fraction

0.5 % at MPP

LID - Light Induced Degradation

Loss Fraction

1.5 %

IAM loss factor

Incidence effect (IAM): User defined profile

0°	30°	45°	60°	70°	75°	80°	85°	90°
1.000	1.000	1.000	0.974	0.907	0.832	0.688	0.445	0.000



Main results

System Production

Produced Energy

32714 kWh/year

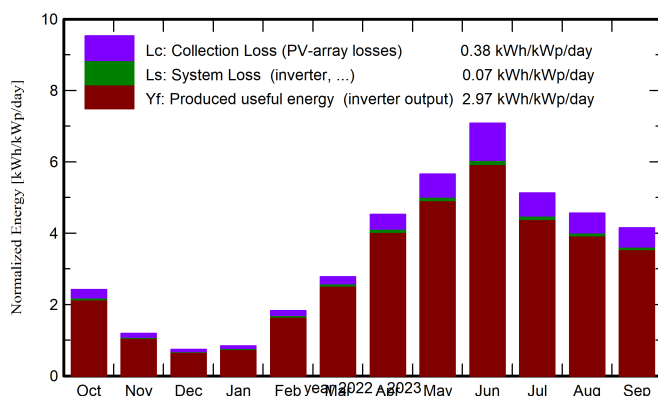
Specific production

1083 kWh/kWp/year

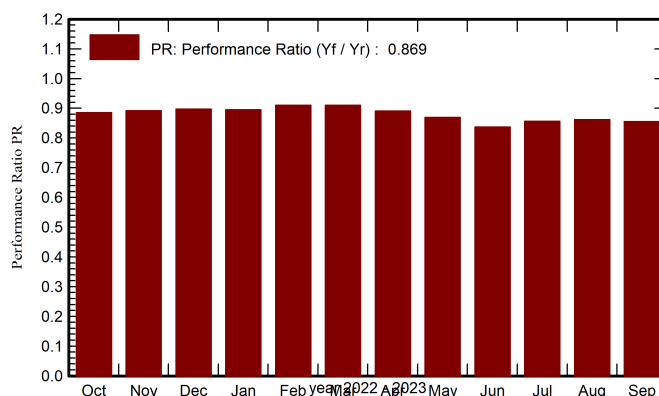
Perf. Ratio PR

86.94 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
Oct. 22	58.6	34.35	12.57	75.0	73.3	2051	2005	0.885
Nov. 22	26.2	17.97	7.86	35.8	34.6	990	964	0.892
Dec. 22	15.4	10.59	3.17	23.0	22.1	642	623	0.897
Jan. 23	19.0	13.97	5.20	26.1	25.2	726	705	0.895
Feb. 23	38.5	22.73	5.09	51.0	49.6	1437	1403	0.911
Mar. 23	72.6	44.34	6.10	86.1	84.2	2420	2367	0.910
Apr. 23	120.5	57.14	8.07	135.9	133.2	3732	3654	0.890
May 23	165.4	75.62	12.35	175.4	172.1	4702	4605	0.869
June 23	202.6	68.93	18.04	212.5	208.9	5486	5374	0.837
July 23	153.4	86.95	17.46	159.0	155.7	4204	4116	0.857
Aug. 23	130.2	73.15	16.76	141.4	138.5	3762	3683	0.862
Sep. 23	104.9	49.11	16.95	124.4	122.0	3282	3214	0.855
Year	1107.3	554.85	10.83	1245.5	1219.5	33434	32714	0.869

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

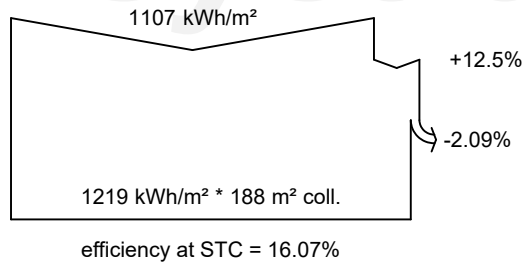
EArray Effective energy at the output of the array

E_Grid Energy injected into grid

PR Performance Ratio



Loss diagram



Global horizontal irradiation

Global incident in coll. plane

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Optimizer efficiency loss

Module quality loss

LID - Light induced degradation

Module array mismatch loss

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

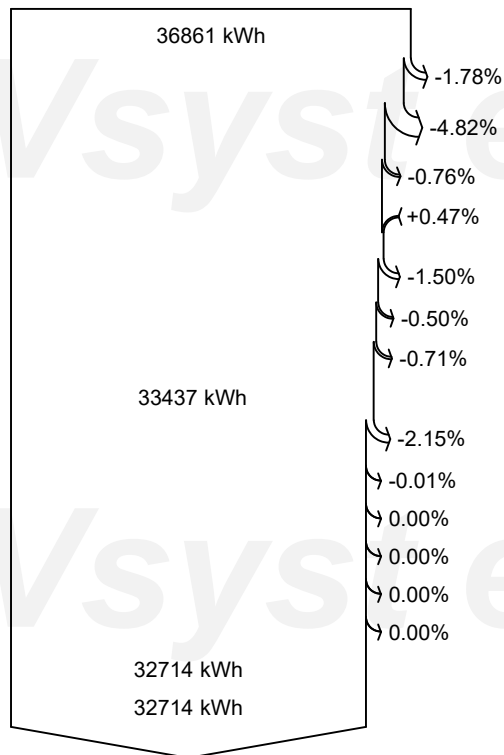
Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Available Energy at Inverter Output

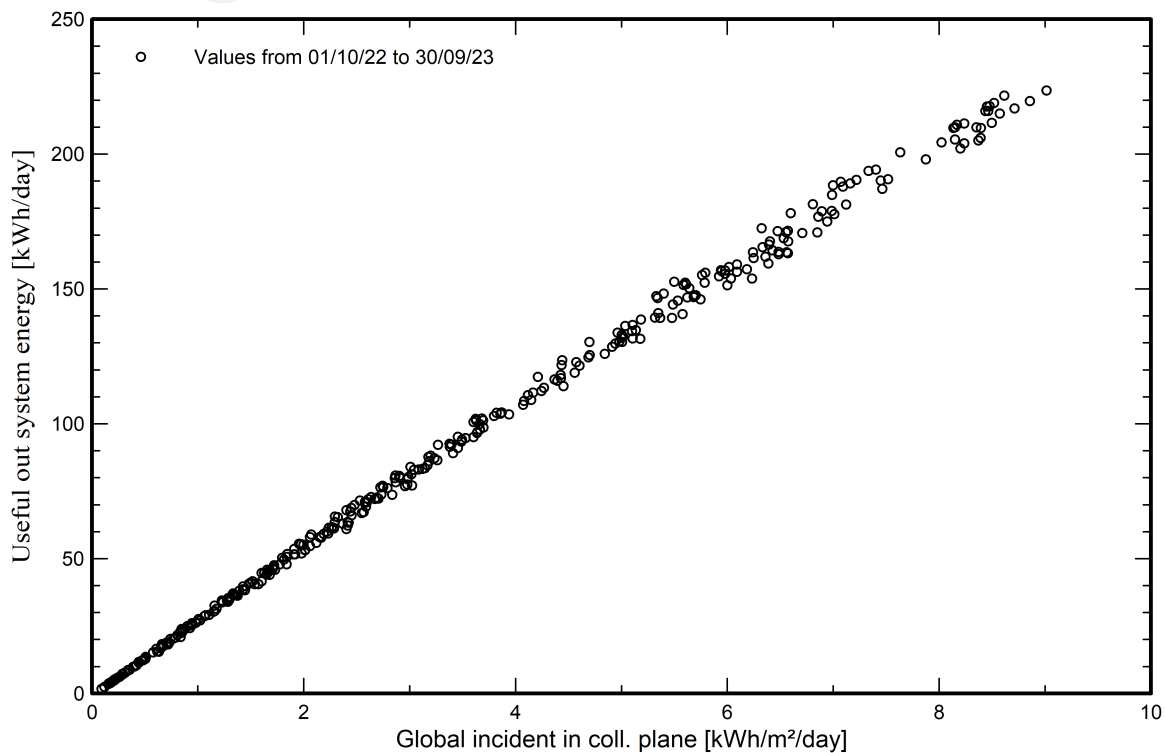
Energy injected into grid





Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

